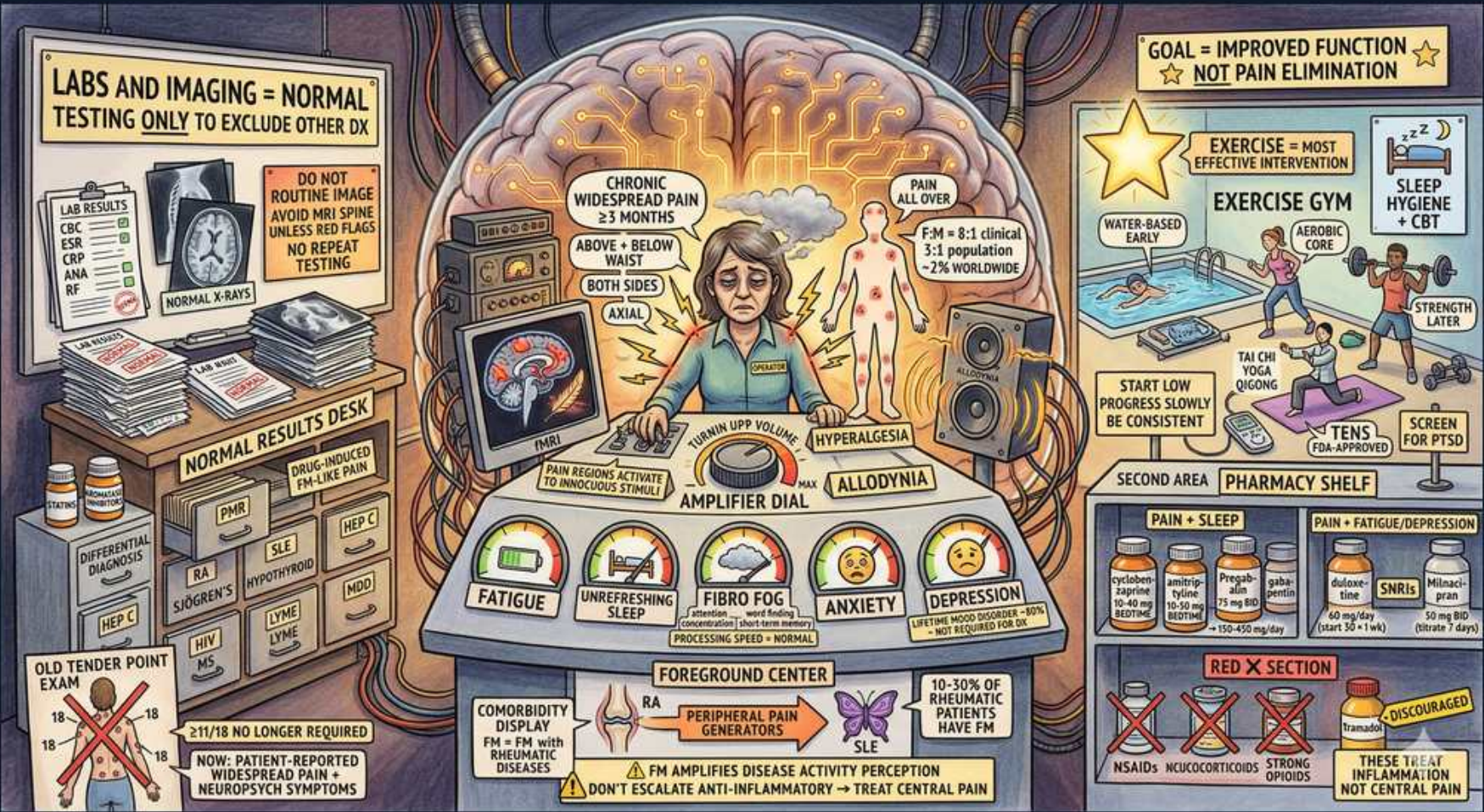


Fibromyalgia — The Pain Amplifier Control Room



1. Chronic widespread pain ≥3 months

WHAT YOU SEE

Central head, 'CHRONIC WIDESPREAD PAIN ≥3 MONTHS', above & below waist, both sides

WHAT IT ENCODES

Defining feature: chronic WIDESPREAD pain present ≥3 months, involving all 4 quadrants (above and below the waist, both sides of the body) plus the axial skeleton. The 2016 ACR criteria operationalize this with a Widespread Pain Index (WPI, 0-19 body regions) plus a Symptom Severity (SS) scale; diagnosis requires WPI ≥7 and SS ≥5 (or WPI 4-6 and SS ≥9). Female predominance (~2:1 to 9:1 in older data), prevalence ~2% of the population.

BOARD TRAP

WRONG answer: diagnosing fibromyalgia from localized or short-duration pain (e.g., one painful shoulder for 2 weeks). The criteria demand WIDESPREAD pain (multiple regions, both sides, above/below waist) for ≥3 months — regional pain syndromes (bursitis, tendinopathy) are the better answer for focal pain.

2. Central amplifier dial

WHAT YOU SEE

Big AMPLIFIER DIAL, 'central pain processing'

WHAT IT ENCODES

Mechanism is central sensitization — amplified CNS pain processing with elevated excitatory neurotransmitters (substance P, glutamate) and reduced descending inhibition (low serotonin/norepinephrine), causing the dorsal-horn 'volume knob' to be turned up. It is a disorder of pain perception, not peripheral tissue inflammation or damage — which is why anti-inflammatory drugs fail and centrally-acting agents (SNRIs, gabapentinoids, TCAs) work.

BOARD TRAP

WRONG answer: ordering joint aspiration, autoantibody panels, or anti-inflammatory therapy looking for peripheral inflammation. Fibromyalgia is CENTRAL pain amplification — there is no synovitis or peripheral inflammatory target, so the answer is to address central sensitization, not to escalate anti-inflammatory work-up/treatment.

3. Hyperalgesia & allodynia

WHAT YOU SEE

Dials labeled HYPERALGESIA and ALLODYNIA

WHAT IT ENCODES

Hyperalgesia (exaggerated pain response to a painful stimulus) and allodynia (pain from a normally non-painful stimulus, e.g., light touch or clothing) are the clinical signatures of central sensitization. They reflect a lowered pain threshold throughout the nervous system rather than localized tissue pathology, and explain why patients hurt 'all over' despite normal exam, labs, and imaging.

BOARD TRAP

WRONG answer: interpreting diffuse tenderness/allodynia as evidence of inflammatory arthritis or myositis. Allodynia and hyperalgesia are hallmarks of central sensitization — a normal CK, ESR/CRP, and joint exam should redirect you to fibromyalgia rather than a myopathy or arthritis.

4. Fatigue

WHAT YOU SEE

Processing gauge 'FATIGUE'

WHAT IT ENCODES

Profound, persistent fatigue is a core symptom alongside pain and is explicitly scored on the Symptom Severity scale of the 2016 ACR criteria (along with unrefreshing sleep and cognitive symptoms). It is typically not relieved by rest and contributes heavily to the functional impairment.

BOARD TRAP

WRONG answer: working up isolated chronic fatigue as primary hypothyroidism, anemia, or chronic fatigue syndrome while ignoring the widespread pain. In fibromyalgia fatigue travels WITH widespread pain and the other core symptoms — but still check TSH and CBC, since hypothyroidism and anemia are key mimics to exclude.

5. Unrefreshing sleep

WHAT YOU SEE

Gauge 'UNREFRESHING SLEEP'

WHAT IT ENCODES

Non-restorative, unrefreshing sleep is characteristic — patients wake unrefreshed regardless of hours slept, classically with alpha-wave intrusion into non-REM (alpha-delta) sleep. Poor sleep amplifies central pain, creating a vicious cycle, which is why low-dose amitriptyline at night and sleep hygiene/CBT are leveraged to improve both sleep and pain.

BOARD TRAP

WRONG answer: ignoring sleep and treating pain in isolation. Unrefreshing sleep is part of the syndrome and a therapeutic lever — addressing it (sleep hygiene, CBT, bedtime amitriptyline) improves pain; also screen for obstructive sleep apnea as a treatable contributor/mimic.

6. Fibro fog (cognitive)

WHAT YOU SEE

Gauge 'FIBRO FOG'

WHAT IT ENCODES

Cognitive dysfunction ('fibro fog') — impaired concentration, word-finding, and working memory — is one of the three core somatic symptoms scored on the Symptom Severity scale (with fatigue and unrefreshing sleep). It fluctuates with pain and sleep and is intrinsic to the syndrome rather than a separate neurologic disease.

BOARD TRAP

WRONG answer: launching an extensive dementia/MS work-up (MRI, LP) for fog in a young patient with widespread pain and normal exam. Fibro fog is part of fibromyalgia's symptom complex — reserve neuro work-up for focal deficits or red flags, not for the typical fluctuating cognitive symptoms.

7. Anxiety / depression comorbidity

WHAT YOU SEE

Gauges 'ANXIETY' and 'DEPRESSION', comorbidity display

WHAT IT ENCODES

Mood disorders — depression and anxiety — are highly prevalent comorbidities (affecting roughly a third or more of patients) and share serotonergic/noradrenergic dysregulation with the pain pathway. This overlap is therapeutically useful: SNRIs (duloxetine, milnacipran) and TCAs (amitriptyline) treat the central pain AND the mood/sleep symptoms simultaneously.

BOARD TRAP

WRONG answer: dismissing fibromyalgia as 'just depression' or somatization. Psychiatric comorbidity is frequent but fibromyalgia is a genuine central pain-processing disorder — treat both; choosing a centrally-acting agent (duloxetine) that covers pain, mood, and sleep is higher-yield than an antidepressant aimed at mood alone.

8. Labs/imaging normal — exclude other dx only

WHAT YOU SEE

Left desk 'LABS AND IMAGING = NORMAL', 'testing ONLY to exclude other dx', normal X-ray/CBC/ESR

WHAT IT ENCODES

Inflammatory markers (ESR/CRP), CBC, CK, and imaging are NORMAL in fibromyalgia; it is a clinical diagnosis (often of exclusion). A focused, limited work-up — TSH, CBC, ESR/CRP, ± CK and vitamin D — is used ONLY to exclude mimics (hypothyroidism, anemia, RA, PMR, polymyositis, hypovitaminosis D). Routine ANA/RF testing is discouraged unless specific features suggest connective tissue disease, because of false positives.

BOARD TRAP

WRONG answer: ordering a broad autoantibody panel (ANA, RF, anti-CCP) as a screen — positive low-titer results in this low-pretest-probability setting lead to misdiagnosis. Normal labs are EXPECTED in fibromyalgia; an elevated ESR/CRP is the red flag that should redirect you to PMR, RA, or another inflammatory disease.

9. No routine/repeat imaging

WHAT YOU SEE

'DO NOT routine image', 'avoid', 'no repeat testing'

WHAT IT ENCODES

Once mimics have been reasonably excluded, avoid repetitive testing and imaging — it does not change management, drives up cost, and amplifies illness anxiety and the central pain cycle. This is a Choosing-Wisely principle: don't keep imaging chronic widespread pain without new localizing red flags.

BOARD TRAP

WRONG answer: re-imaging or re-running labs each visit to 'find the cause.' Over-investigation is a pitfall that reinforces the patient's symptom focus — after an appropriate initial exclusion work-up, the answer is to pivot to treatment (exercise, CBT, centrally-acting drugs), not more tests.

10. Tender point exam no longer required

WHAT YOU SEE

Bottom-left 'OLD TENDER POINT EXAM' with X, '11/18 no longer required'

WHAT IT ENCODES

The old 1990 ACR criterion of ≥11 of 18 tender points on digital palpation is no longer required. Modern criteria (2010/2011/2016 ACR) are patient-reported: the Widespread Pain Index (count of painful body regions) plus the Symptom Severity scale (fatigue, unrefreshing sleep, cognitive symptoms). This shift recognizes fibromyalgia as a central symptom complex rather than a count of tender spots.

BOARD TRAP

WRONG answer: excluding fibromyalgia because the patient has fewer than 11 tender points. The tender-point exam is outdated and not required — diagnosis now rests on WPI + Symptom Severity, so a low tender-point count does not rule it out.

11. Exercise = most effective treatment

WHAT YOU SEE

Exercise gym, 'EXERCISE = MOST EFFECTIVE INTERVENTION', start low/go slow, CBT, sleep hygiene

WHAT IT ENCODES

Graded aerobic exercise is the single most effective intervention and the cornerstone of management (strongest evidence of any single modality); low-impact options like walking, swimming/water-based exercise, and tai chi/yoga work well. Combine with patient education, CBT, and sleep hygiene. Start low and progress slowly to avoid post-exertional flares and improve adherence.

BOARD TRAP

WRONG answer: making a drug the first-line treatment. Non-pharmacologic therapy — graded aerobic exercise plus CBT and education — is first-line and the most effective intervention; medications are adjuncts added when non-drug measures are insufficient, not the opening move.

12. Duloxetine / pregabalin / amitriptyline

WHAT YOU SEE

Pharmacy shelf: duloxetine, pregabalin, amitriptyline; pain–sleep / pain–depression

WHAT IT ENCODES

Effective centrally-acting pharmacotherapy (3 are FDA-approved: duloxetine, milnacipran, pregabalin): SNRIs duloxetine (30→60 mg/day) and milnacipran; the gabapentinoid pregabalin (start 75 mg BID, titrate to 150-225 mg BID; gabapentin off-label); and low-dose TCA amitriptyline (10-25 mg at bedtime). Choose by symptom cluster — duloxetine/milnacipran if depression/fatigue dominate; pregabalin or bedtime amitriptyline if sleep disturbance dominates.

BOARD TRAP

WRONG answer: choosing an NSAID, opioid, or oral steroid as the pharmacologic agent. The drug answer is a centrally-acting agent (duloxetine, milnacipran, pregabalin, or low-dose amitriptyline) targeting pain plus sleep/mood — peripheral anti-inflammatory and opioid analgesics do not work for central sensitization.

13. AVOID opioids, NSAIDs, glucocorticoids

WHAT YOU SEE

Red 'RED □ SECTION': opioids, NSAIDs, steroids crossed out, 'discouraged'

WHAT IT ENCODES

Opioids should be avoided/are contraindicated — they are ineffective for central pain, carry dependence/overdose risk, and can paradoxically worsen pain via opioid-induced hyperalgesia. NSAIDs and glucocorticoids are ineffective because there is no peripheral inflammation to suppress. (Tramadol has weak SNRI activity and is sometimes used cautiously, but pure opioids are discouraged.)

BOARD TRAP

WRONG answer (the classic distractor): escalating to opioids, adding chronic NSAIDs, or prescribing glucocorticoids for fibromyalgia pain. There is no inflammation to treat, and opioids worsen central pain (hyperalgesia) — the correct choices are exercise/CBT and centrally-acting agents.

14. Don't escalate anti-inflammatories

WHAT YOU SEE

Foreground 'FM amplifies disease activity perception — treat CENTRAL pain', 'don't escalate anti-inflammatory'

WHAT IT ENCODES

Fibromyalgia commonly coexists with RA, SLE, and other rheumatic diseases and inflates patient-reported disease-activity scores (e.g., tender joint counts, DAS28) without true inflammatory worsening. When objective markers (ESR/CRP, swollen joints, imaging) are quiet but the patient reports high pain, treat the central pain rather than reflexively escalating DMARDs, biologics, or steroids.

BOARD TRAP

WRONG answer: intensifying immunosuppression (more DMARDs/biologics/steroids) because an RA or lupus patient reports worsening pain, when inflammatory markers and joint exam are normal. The discriminator is objective inflammation — if it's absent, the pain is overlapping fibromyalgia (central sensitization), which is treated with exercise/CBT and centrally-acting agents, not more immunosuppression.
